

ROOM: A Human–Humanoid Interaction Manipulation Dataset for Nonverbal Behavior Grounding

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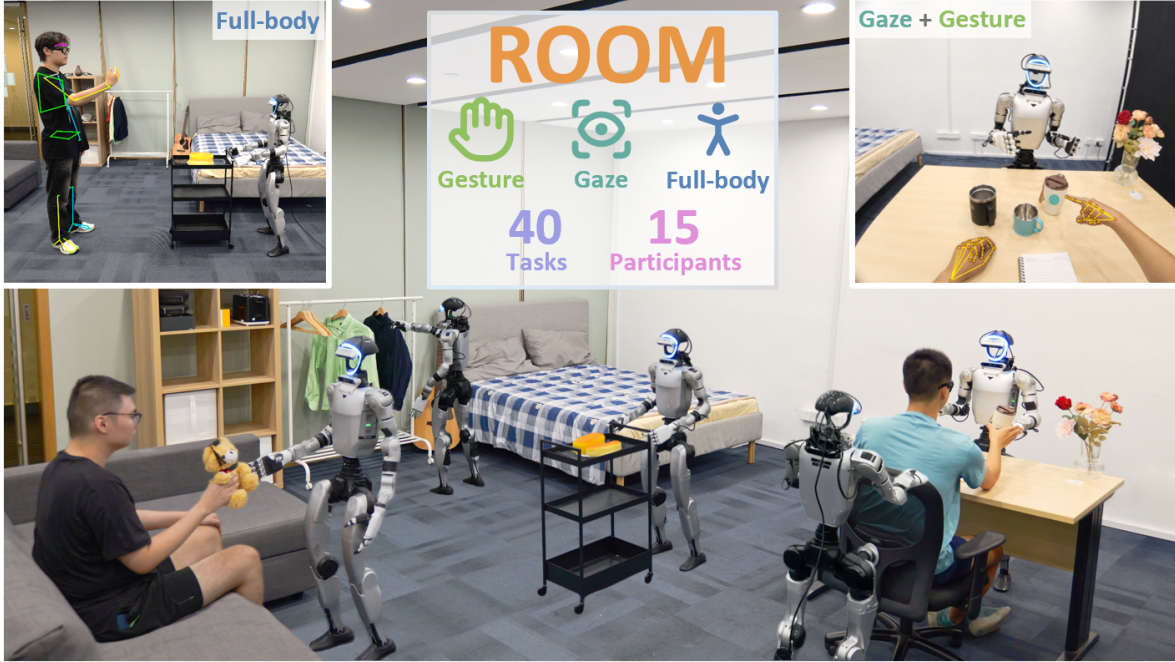


Fig. 1. Synchronized human nonverbal behavior and humanoid manipulation in ROOM.

Abstract—Humanoid robots can perform increasingly complex manipulation tasks, but still struggle to understand human intentions and adapt their actions during collaboration. This requires identifying which object a person indicates, when help is needed, and when an ongoing action should change. People communicate this information through nonverbal behavior, including gaze, gestures, and body posture, alongside spoken language. Existing manipulation datasets rarely capture these behaviors together with the corresponding robot responses. We introduce Referential and Ongoing-interaction Observations for Manipulation (ROOM), a dataset and benchmark for nonverbal behavior grounding in human–humanoid manipulation. ROOM contains 2.5k demonstrations from 12 participants across 40 HRI tasks spanning tabletop manipulation and locomotion, recording robot manipulation data and human nonverbal behavior. The tasks cover referential grounding and interaction state grounding, requiring robots to select targets, determine when to act, and respond to feedback based on changes in human behavior. Real-robot evaluations of three representative vision–language–action policies show that training on ROOM improves their ability to use human nonverbal information to guide actions, increasing task success and reducing grounding failures. Comparisons under a common backbone show that visual prompts improve target selection, while semantic descriptions improve response selection and action timing. A 12-participant user study further shows that nonverbal interaction improves task success and reduces perceived workload compared with real-time verbal guidance. ROOM supports learning and evaluating how humanoids adapt their actions to human nonverbal behavior.

I. INTRODUCTION

Humanoid robots are emerging as an important platform for embodied intelligence, as their human-like morphology allows them to operate in environments designed around human physical capabilities. Recent advances in whole-body control and robot learning have expanded their capabilities, enabling them to perform increasingly complex tasks that integrate locomotion and dexterous manipulation [1]. These capabilities have been demonstrated in everyday human environments, including human–humanoid interaction scenarios [2]. However, effective collaboration with humans in such settings requires more than autonomous task execution: humanoids must infer their human partners’ goals from the unfolding interaction and adapt their actions accordingly [3]. This raises a central question: *can humanoids read the room?*

Human–human interaction provides a natural model for this capability. Human collaboration is commonly studied as a form of joint action, in which partners coordinate their behavior toward shared goals [4]. Classical accounts of grounding in communication suggest that successful collaboration requires the coordination of both content and process: partners need to maintain a common understanding of what they are jointly doing and how the joint activity is unfolding [5]. Importantly, such coordination does not rely on language alone. In face-to-face interaction, nonverbal

behavior such as gaze, hand gestures, body posture, and movement forms an integral part of this coordination, complementing spoken language in conveying both the content of the joint activity and how that activity is progressing [6]. As shown in Fig. 1, gaze and gestures can indicate what the interaction is about, while body posture and movement can indicate how it should proceed [7], [8]. Together, these nonverbal cues provide essential information for interpreting a partner’s intent and determining an appropriate response.

Realizing such nonverbal understanding in robot learning policies depends critically on the data used for training and evaluation. Large-scale datasets have driven recent progress in robot manipulation by covering diverse tasks, environments, and robot embodiments [9], [10], [11], [12], but they contain little information about the human partner and their behavioral signals. More recently, humanoid and human-present datasets have begun to incorporate human–robot interaction into manipulation tasks [2], [13]. However, they are not designed to systematically study nonverbal behavior grounding. Human-present datasets typically focus on specific cues or interaction functions, while humanoid datasets emphasize broad manipulation coverage rather than rich multimodal human behavior. As a result, real-world human–humanoid datasets that jointly capture multiple nonverbal signals, while relating variations in these signals to different robot actions under matched task contexts, remain scarce.

To fill this gap, we introduce **ROOM**, a dataset and benchmark for nonverbal behavior grounding in human–humanoid manipulation. ROOM is structured around interaction contexts in which variations in human nonverbal behavior call for different robot action decisions. ROOM comprises 40 HRI manipulation tasks spanning tabletop manipulation and whole-body collaboration. Following the distinction between coordination of content and process in collaborative activity [5], ROOM is organized around two forms of nonverbal grounding: *referential grounding* and *interaction-state grounding*. Together, these tasks require a humanoid to determine what a person refers to, what response is needed, when to act, and whether to revise an ongoing action. ROOM contains 2.5k demonstrations from 15 participants. Each demonstration aligns robot-view observations and manipulation trajectories with human egocentric video, gaze, hand motion, and full-body pose.

Our main contributions are as follows. (1) We introduce ROOM, a dataset and benchmark for nonverbal behavior grounding in human–humanoid interaction, containing 40 HRI tasks spanning tabletop manipulation and locomanipulation. ROOM covers referential grounding and interaction state grounding, with synchronized robot trajectories and multimodal human behavior. (2) We benchmark representative VLA policies across eight evaluation conditions spanning four task groups and distinguish nonverbal behavior grounding failures from physical execution failures. We show that pretraining on ROOM improves task success and reduces nonverbal behavior grounding failures in the evaluated tasks. (3) We systematically compare nonverbal information representations and integration strategies under a common policy

backbone, examining their effects on robot action decisions and task success. (4) We conduct a user study using NASA-TLX, showing that nonverbal interaction reduces perceived workload compared with detailed verbal guidance.

II. RELATED WORK

A. Human-Aware Manipulation Datasets

Large-scale manipulation datasets such as Open X-Embodiment, DROID, RoboMIND, and AgiBot World scale policy learning across embodiments, tasks, and environments [9], [10], [11], [12]. Humanoid Everyday extends this line to full-body humanoids and includes basic human–robot interaction tasks [2]. HARMONIC records gaze, body pose, human egocentric video, user control, and robot state for intention modeling in assistive shared autonomy [14], but does not pair robot-view observations with manipulation trajectories for end-to-end policy learning and thus falls outside the manipulation-policy datasets in Table I. HABIT collects human-present bimanual demonstrations and evaluates synchronization, yielding, and gesture following [13]. ROOM jointly records gaze, gesture, body motion, human egocentric observations, and humanoid manipulation trajectories to study how nonverbal information shapes action decisions across referential and interaction state grounding.

B. Human-Aware Manipulation Policies

Human–robot interaction has long studied how robots can adapt manipulation to human behavior. Traditional human-aware manipulation methods either infer human goals and adaptation to select a collaboration strategy or assistance [15], [16], or coordinate robot manipulation from human actions and task progress [17]. These approaches use structured, task-specific interaction models to relate human behavior to robot responses. Recent end-to-end policies instead condition action generation on *human-present demonstrations* [13] or explicit representations of human behavior. At the perception level, spatial indications are incorporated as visual prompts [18], [19], [20]; at the semantic level, human behavior is converted into descriptions combined with task instructions [20]; and at the action level, spatial information directly conditions action prediction [19]. Together, these studies show how human behavior can influence robot actions through human-present observations or explicit conditioning at different levels. ROOM complements them by evaluating whether existing VLA policies ground nonverbal information from raw visual observations and by comparing representation–integration mechanisms under the same backbone and task setting.

III. THE ROOM DATASET

We intend ROOM to support the development and evaluation of robot policy that use human nonverbal information to select target objects or locations, determine when to act, and respond to human feedback. In this section, we summarize the dataset and interaction task design, followed by the data collection and processing procedures.

TABLE I
COMPARISON OF ROBOT MANIPULATION DATASETS WITH EMBODIED HRI SIGNALS.

Dataset	Embodiment	HRI manipulation	HRI loco-manip.	Gaze	Gesture	Body pose	Human ego-view	Nonverbal grounding
Open X-Embodiment [9]	Multi-embodiment	–	–	–	–	–	–	–
DROID [10]	Arm	–	–	–	–	–	–	–
RoboMIND [11]	Multi-embodiment	–	–	–	–	–	–	–
AgiBot World [12]	Wheeled dual arm	Partial	–	–	–	–	–	–
Humanoid Everyday [2]	Humanoid	Partial	–	–	–	–	–	–
HABIT [13]	Dual arms	✓	–	–	✓	–	✓	Gesture Partial
ROOM (Ours)	Humanoid	✓	✓	✓	✓	✓	✓	✓

A. The ROOM Dataset

The ROOM dataset contains 40 HRI tasks spanning tabletop manipulation and loco-manipulation, with 2.5k demonstrations collected from 12 participants. Each demonstration records robot-view observations and humanoid action trajectories synchronized with human egocentric video and human nonverbal information, including gestures, body posture, and gaze.

B. Interaction Task Design

ROOM task design draws on the distinction between coordinating the content and process of joint action [5]. Human nonverbal behavior contributes to both by indicating referents and helping partners coordinate actions and provide feedback [8], [21]. For human–humanoid interaction manipulation, we focus on target selection within content coordination and action adjustment within process coordination. These motivate the two grounding requirements examined in ROOM, *referential grounding* and *interaction state grounding*. We design tasks in which the language instruction alone does not fully specify the intended target or response, requiring the humanoid to use human nonverbal information to determine the appropriate action.

We characterize ROOM tasks by the nonverbal grounding requirements they impose. *Referent identification* tasks assess referential grounding by requiring the humanoid to identify intended objects or locations from gaze or pointing [22], [8]. *Interaction-state grounding* is instantiated through three interaction functions. *Anticipatory assistance* requires the humanoid to observe the person’s ongoing activity and body posture, infer what help is needed next, and take the corresponding assistive action [23], [24]. *Action coordination* requires the humanoid to determine when to act, wait, or resume during collaboration based on the person’s progress and readiness [4], [25]. *Feedback and steering* requires the humanoid to interpret human nonverbal feedback and determine whether to continue or how to adjust an ongoing action [21]. Individual tasks may involve more than one of these grounding requirements.

C. Data Collection

a) Collection Setup: We collected ROOM in a home environment using a Unitree G1 humanoid equipped with Inspire dexterous hands, as shown in Fig. 2. Intel RealSense D455 cameras mounted on the humanoid’s head and chest recorded RGB observations. The head camera was tilted downward to capture the manipulation workspace, while the chest camera mainly captured the participant’s gestures and body posture. Participants wore Project Aria glasses to record

egocentric video and obtain gaze and hand-motion information. A markerless multi-camera motion-capture system was deployed around the workspace to capture full-body motion. The system has a reported mean 3D landmark error of 13.23 mm on RRIS40 [26].

b) Interaction Protocol: Each task was specified by an interaction scenario and a language instruction. A participant performed the human role, while a separate operator teleoperated the humanoid to execute the intended robot response. Across demonstrations, we varied the interacting participant and clothing, as well as the categories and positions of the manipulated objects.

c) Robot Teleoperation: We used the SONIC whole-body controller [27] to collect loco-manipulation demonstrations. The operator controlled whole-body motion using a PICO VR headset, handheld controllers, and two ankle trackers. For tabletop tasks, we used the Meta Quest 3 hand-tracking interface to capture finer-grained hand motion. The operator’s arm and hand targets were retargeted to the humanoid through inverse kinematics.

D. Data Processing

We processed the recorded sensor streams to extract human nonverbal information and align it with robot-view observations and action trajectories. Project Aria recordings were processed using Machine Perception Services (MPS) to obtain timestamped gaze directions and 21-point hand landmarks [28]. The multi-camera motion-capture system tracked the global positions of the participant, teleoperator, and robot and provided full-body joint trajectories. Its multiple viewpoints helped recover body joints occluded in the robot’s onboard views.

To support policies that rely only on onboard RGB observations, we additionally processed images from the robot’s head and chest cameras using off-the-shelf hand and human pose estimation methods [29], [30]. We extracted 2D hand and body keypoints from the robot-view images. We synchronized the extracted human nonverbal information with robot-view observations and action trajectories at 30 Hz and organized the demonstrations in the LeRobot dataset format.

IV. NONVERBAL INFORMATION FOR VLA POLICIES

To study how VLA policies use human nonverbal information, we compare prompted image tokens, spatiotemporal behavior tokens, and semantic language tokens under a common GR00T N1.7 backbone. Fig. 3 summarizes these representations and their integration mechanisms.

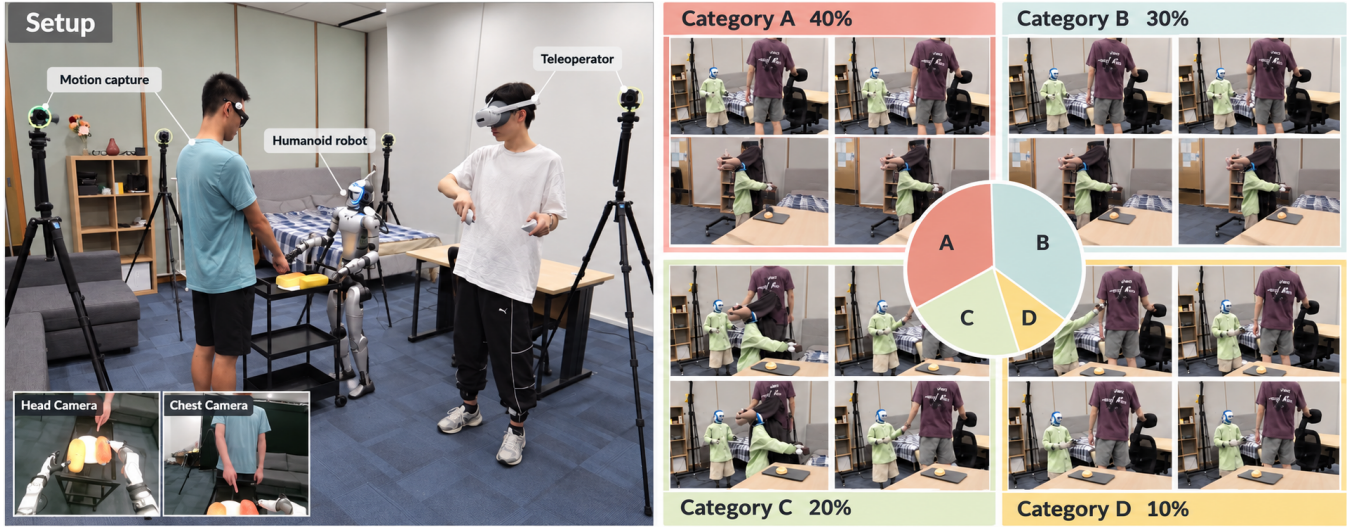


Fig. 2. Draft Version.ROOM data collection setup. A participant wearing Project Aria glasses interacts with a teleoperated Unitree G1 humanoid, while a markerless multi-camera system captures full-body motion. The bottom insets show the robot’s head-camera and chest-camera views and the participant’s egocentric view.

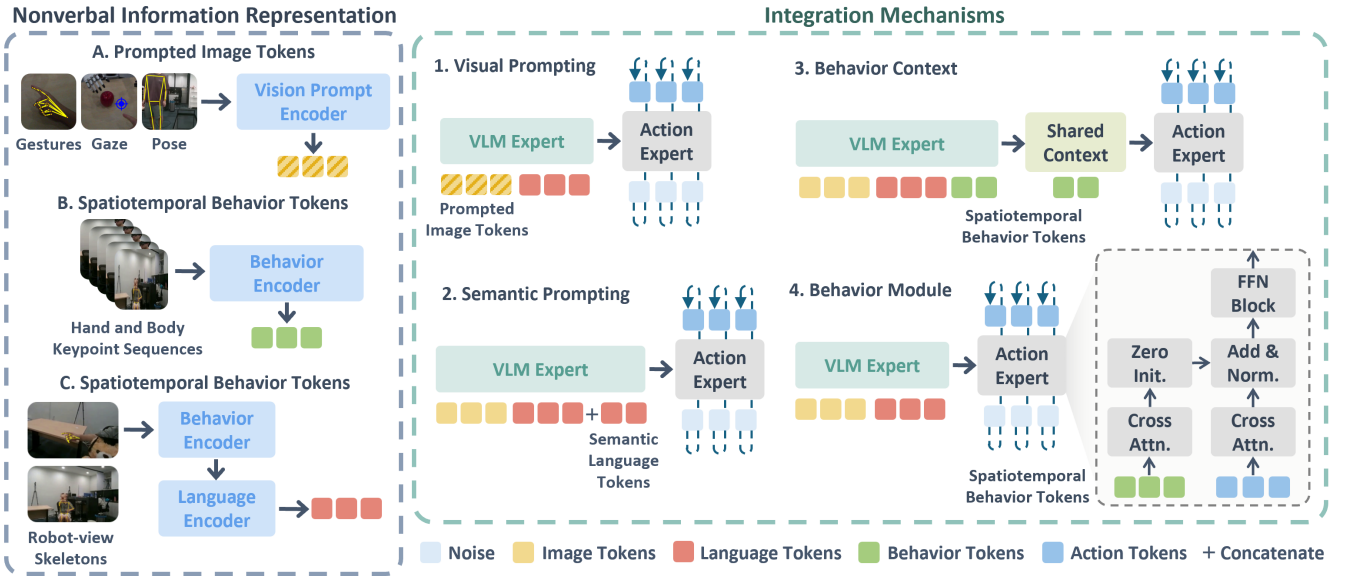


Fig. 3. Nonverbal behavior representations and their integration into VLA policies.

A. Preliminaries

GR00T N1.7 consists of a vision–language backbone that processes visual and language inputs and a flow-matching DiT action expert for action generation [31]. Within the DiT, self-attention models relationships among robot-state and action tokens, while cross-attention allows these tokens to attend to the VLM’s output features.

B. Nonverbal Information Representations

a) *Prompted Image Tokens*: We overlay hand landmarks and body-pose keypoints on robot-view RGB observations to make gestures and body movements explicit while retaining the surrounding scene. For referential grounding, we extend a line from the index fingertip along the estimated pointing direction to highlight the likely target object. Gaze points are visualized in the participant’s egocentric images. The VLM’s visual encoder converts the augmented images into prompted image tokens [19], [20].

b) *Spatiotemporal Behavior Tokens*: We represent recent human behavior using hand and body keypoint sequences extracted from robot-view RGB observations. Keypoints are sampled at four-frame intervals within a short temporal window to reduce temporal redundancy while capturing changes in hand configuration and body posture. A two-layer MLP encodes the sampled sequence into behavior tokens.

c) *Semantic Language Tokens*: We express recognized gestures and body postures as text [20]. A lightweight classifier predicts behavior labels from robot-view skeletons. These labels are converted into descriptions such as “human is pointing” or “human is sitting down” and appended to the task instruction as supplementary language prompts. Because occlusion can leave robot-view skeletons incomplete, we train the classifier on these skeletons using body-posture labels derived from the more complete skeletons reconstructed by the markerless multi-camera system. At inference, the classifier uses only skeletons extracted from the robot’s head

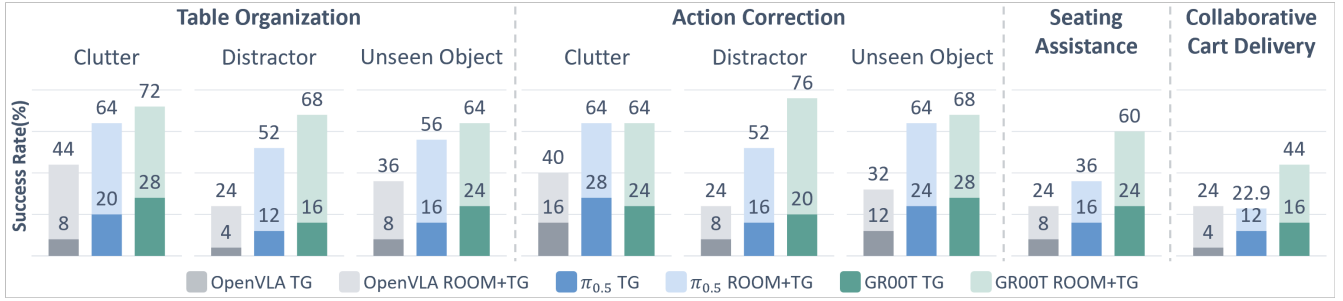


Fig. 4. Nonverbal HRI benchmark across eight evaluation conditions. Bars show task success rates (%). TG denotes task-group training, and ROOM + TG denotes pretraining on all 40 ROOM tasks followed by the same task-group training.

and chest cameras.

C. Integration Mechanisms

a) *Visual Prompting*: Visual Prompting introduces the augmented robot-view images through GR00T’s existing visual input, highlighting gestures, body posture, and pointing directions while retaining the surrounding scene. The VLM processes these images together with the task instruction and supplies the resulting features to the action expert without architectural changes. We also evaluate gaze-based visual prompting in the gaze and gesture ablation in Sec. V-F.

b) *Semantic Prompting*: Semantic Prompting appends descriptions of the recognized gestures and body postures to the task instruction and updates them as the participant’s behavior changes. GR00T processes the combined text through its existing language pathway, allowing the policy to use the person’s current behavior to determine when an action should begin, continue, or change.

c) *Behavior Context*: Behavior Context integrates the same behavior tokens at either the VLM input or output after projecting them to the required feature dimension. At the VLM input, the tokens are concatenated with the image and language tokens for joint processing by the VLM. At the VLM output, they are concatenated with the vision and language features and passed to the action expert through its existing cross-attention pathway.

d) *Behavior Module*: Behavior Module adds a cross-attention branch to the action expert, separate from the existing VLM conditioning. Action features serve as queries, while behavior tokens provide keys and values. The branch output passes through a zero-initialized projection and is added to the original action features. This preserves the original features at initialization while allowing behavior-dependent adjustments during fine-tuning [19].

V. EXPERIMENTS

We use ROOM to evaluate the nonverbal HRI manipulation capabilities of existing VLA policies and to study how explicit nonverbal information affects policy behavior and human interaction outcomes. The experiments address four questions:

- 1) How do existing VLA policies perform on referential grounding and interaction state grounding tasks, and does ROOM pretraining followed by task-group training improve performance?

- 2) How often do failures arise from referential grounding, interaction state grounding, or physical execution?
- 3) How do different nonverbal representations and integration mechanisms affect nonverbal behavior grounding failures and task success?
- 4) Compared with detailed language alone, can nonverbal information reduce the required language specificity, human workload?

A. Experimental Setup

a) *Experimental tasks*: We select four task groups from ROOM, covering two tabletop and two whole-body interaction settings. The tabletop tasks are **Table Organization**, which involves *referent identification*, and **Action Correction**, which involves both *referent identification* and *feedback and steering*. The whole-body tasks are **Seating Assistance**, which involves *anticipatory assistance*, and **Collaborative Cart Delivery**, which involves *action coordination*. For the two tabletop task groups, we further evaluate three scene conditions. *Distractors* introduces objects of the same type or with similar appearance; *Unseen Objects* uses target objects absent from the training demonstrations; and *Clutter* increases the number of tabletop objects to three to five.

Across these tasks, the main experiments focus on gestures and body posture, which occur across both referential and interaction-state interactions. Gaze primarily supports target selection in our task design and is evaluated separately as a referential cue in Sec. V-F.

b) *Policy and training*: We evaluate OpenVLA-OFT, $\pi_{0.5}$ [32], and GR00T N1.7 [31] under two training settings. Each task group includes related tasks with different objects or interaction variants. In task-group training (TG), each policy is fine-tuned on the corresponding task group. ROOM + TG first pretrains the policy on all 40 ROOM tasks and then applies the same task-group training. The representation and integration experiments use GR00T N1.7, share the same checkpoint, training episodes, and number of training steps across variants.

c) *Real-robot evaluation*: For each policy, we conduct 25 rollouts per task condition on the Unitree G1, distributed across three participants. During evaluation, we vary the participant and rearrange the objects across rollouts. Within each task condition, the task instruction remains the same, while the participant changes the gesture or body movement to indicate a different target or robot response.

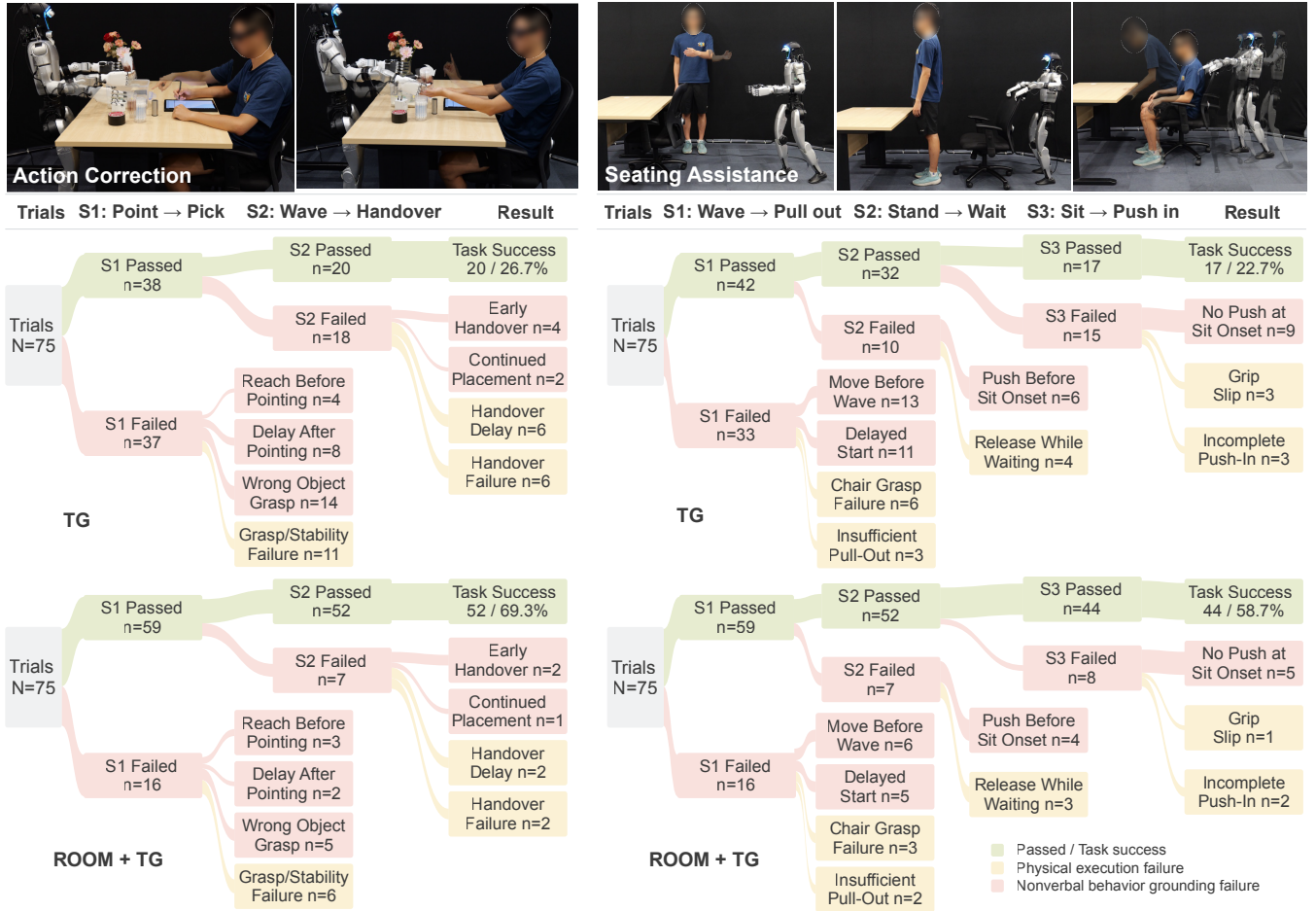


Fig. 5. Stage-wise failure decomposition for Action Correction and Seating Assistance.

To distinguish failures in nonverbal behavior grounding from physical execution failures, we classify each failed rollout according to the type of failure that occurs first: referential grounding, interaction-state grounding, or physical execution. Referential grounding failures indicate incorrect target selection. Interaction state grounding failures indicate an incorrect response or action timing. Physical execution failures occur when the decision is correct but the robot fails to grasp, transport, place, or hand over the object.

B. Nonverbal HRI Benchmark

We evaluate whether training on ROOM helps existing VLA policies use human nonverbal information to guide their actions. We evaluate each policy under two training settings: TG and ROOM + TG. All policies use their standard robot-view observations and task instructions.

Fig. 4 shows that ROOM + TG improves task success across all three policies, with an average gain of 32.6 percentage points over TG. GR00T N1.7 achieves the highest average success rate under ROOM + TG. These results suggest that training on ROOM dataset improves policies’ ability to use human nonverbal information for target selection and action adjustment in response to human feedback.

C. Grounding Failure Analysis

We analyze GR00T N1.7 on two representative multi-stage tasks. In Action Correction task, the participant first points to the target object and then uses a corrective gesture to redirect the robot from object placement to handover. In Seating Assistance, a waving gesture prompts the robot to approach and pull out the chair. The robot then waits until the participant begins to sit, at which point it pushes the chair in to assist seating.

The failure analysis reveals substantial nonverbal behavior grounding failures under TG. As shown in Fig. 5, about two-thirds of the failures in both tasks occur during nonverbal behavior grounding rather than physical execution. Compared with TG, ROOM + TG increases task success rate from 20/75 to 52/75 in Action Correction and from 17/75 to 44/75 in Seating Assistance. In Action Correction, the policy makes fewer wrong-object selections and responds more reliably to pointing and corrective gestures. In Seating Assistance, the policy makes fewer action-timing failures when responding to beckoning gestures and sitting motions. In addition, physical execution failures also decrease in both tasks. Under the same task instruction, ROOM pretraining improves target selection and reduces premature actions and failures to respond to corrective gestures, suggesting benefits for both referential grounding and action timing and

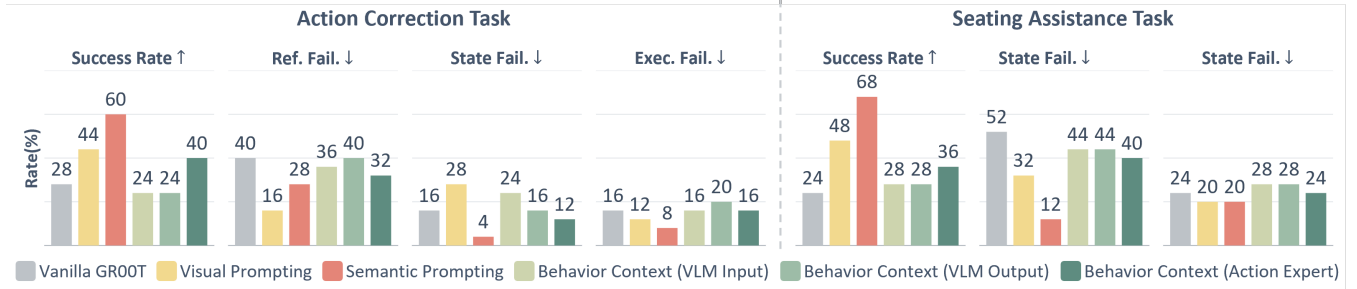


Fig. 6. Nonverbal representations and integration strategies on GROOT N1.7. Rates (%) are based on 25 rollouts per task and variant, with failures classified by the first occurrence. Ref., State, and Exec. denote referential grounding, interaction state grounding, and physical execution failures.

TABLE II

VERBAL AND NONVERBAL COMMUNICATION. BOTH USE SEMANTIC PROMPTING WITH THE SAME GOAL-LEVEL TASK INSTRUCTION.

Communication function	Table Organization			Seating Assistance		
	Decision ↑	Success ↑	TLX ↓	Decision ↑	Success ↑	TLX ↓
Verbal	32/50	26/50	46.7 ± 13.3	28/50	24/50	46.7 ± 13.3
Nonverbal	36/50	32/50	25.0 ± 6.7	40/50	36/50	25.0 ± 6.7

Note. Decision and Success denote action-decision accuracy and task success rate (%). TLX is the unweighted mean of six adapted NASA-TLX dimensions, each rated from 1 to 7 and rescaled to 0–100, with higher scores indicating greater workload. Values are reported as mean ± standard deviation across participants.

TABLE III

GAZE AND GESTURE ABLATION.

Explicit nonverbal cues	Task success ↑	Ref. fail ↓	Exec. fail ↓
None (RGB only)	–	–	–
Gaze only	–	–	–
Gesture only	–	–	–
Gaze + gesture	–	–	–

adjustment during interaction.

D. Nonverbal Representations and Integration

In this section, we investigate how nonverbal information should be represented and integrated into the policy. We compare Prompted Image Tokens, Semantic Language Tokens, and Spatiotemporal Behavior Tokens as described in Sec. IV-C.

Fig. 6 shows that Visual Prompting reduces referential grounding failures in Action Correction from 10/25 to 4/25, while interaction state grounding failures increase from 4/25 to 7/25. We also observe that providing the baseline with more specific language descriptions does not reliably resolve target ambiguity when similar distractors are present. Visual overlays, in contrast, make the spatial relationship between pointing and candidate objects explicit, helping the policy identify the intended target. However, overlays may obscure fine hand details needed to interpret corrective gestures and switch actions. In Seating Assistance, interaction state grounding failures decrease from 13/25 to 8/25, suggesting that whole-body overlays make larger posture changes easier to distinguish.

Semantic Prompting achieves the highest success in both tasks, reaching 15/25 in Action Correction and 17/25 in Seating Assistance, with interaction state grounding failures reduced to 1/25 and 3/25. In Action Correction, advance instructions to follow gestures do not reliably trigger the switch from placement to handover, whereas timely behavior descriptions help. In Seating Assistance, providing “human is starting to sit down” while the participant remains standing

does not immediately trigger chair pushing. The robot waits until the participant begins to sit before pushing the chair forward. These observations suggest that behavior descriptions help specify the appropriate response, while visible human behavior further informs action timing.

Spatiotemporal Behavior Tokens offer little benefit when inserted at the VLM input or output and reduce success in Action Correction. These tokens may be poorly aligned with pretrained features and interfere with existing visual-language processing. Integration at the action expert yields modest improvements but remains less effective than prompting, suggesting that compatibility with the pretrained policy matters more than simply adding behavior features.

E. User Study: Verbal Guidance and Nonverbal Interaction

To assess whether nonverbal communication naturally used in human collaboration reduces coordination burden in human–humanoid interaction, we conduct a user study with 12 participants comparing real-time verbal guidance and nonverbal interaction in Table Organization and Seating Assistance, using adapted NASA-TLX to measure perceived workload. Participants complete the questionnaire after each task under each condition. Under the same goal-level instruction and semantic prompting interface, we additionally collect verbal demonstrations replacing gestures for target selection and body posture for chair-pushing timing. Both policies start from the same ROOM-pretrained checkpoint and use equal demonstration counts and training budgets.

Nonverbal interaction improves action-decision accuracy from 64% to 72% in Table Organization and from 56% to 80% in Seating Assistance. Task success also increases, from 52% to 64% and from 48% to 72%, respectively. NASA-TLX scores decrease from 46.7 ± 13.3 to 25.0 ± 6.7 in both tasks. Under verbal guidance, participants must track the robot’s progress and issue timely action and correction commands, coordinating their movements with the robot’s

execution. With nonverbal interaction, the robot adjusts target selection and action timing to participants' gestures and body movements. Participants can thus communicate through their ongoing actions, reducing the need for continuous monitoring and explicit verbal guidance.

F. Gaze and Gesture Ablation

We examine whether explicit gaze and gesture information provide complementary benefits for referential grounding in *Table Organization*. We compare RGB-only observations with gaze, gesture, and their combination, using the same task instruction, data split, and training budget.

Human behavior remains visible in the RGB-only condition. Comparing the combined input with each modality alone assesses whether gaze provides additional information beyond gestures.

VI. CONCLUSION

We introduced ROOM to make nonverbal interaction a trainable and measurable component of human–humanoid manipulation. Across 40 tabletop and whole-body tasks, ROOM pairs human gaze, gestures, and body motion with the robot responses these behaviors are intended to elicit. The experiments show that task-group training alone leaves substantial referential grounding and interaction state grounding failures, whereas training across ROOM improves both nonverbal behavior grounding and physical execution. They also show that explicit behavior representations and their integration into the policy affect what the robot learns from these signals. The user study further indicates that nonverbal interaction can reduce participant workload without sacrificing task performance. Overall, ROOM shows that learning how human behavior changes the appropriate robot response is a distinct requirement for humanoid interaction, beyond learning how to execute manipulation tasks.

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